Lift-and-Project Hierarchies Collapse under Symmetry

Yuri Faenza[1] Víctor Verdugo[23] José Verschae[3]
Matías Villagra[1]

Berkeley - Columbia Meeting in Engineering and Statistics

October 3rd, 2025

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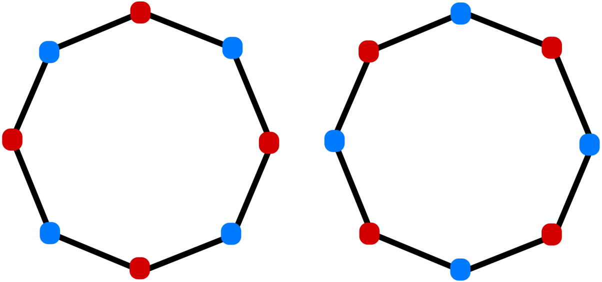
[1] IEOR, Columbia University
[2] DIIS, Pontificia Universidad Católica de Chile
[3] IMC, Pontificia Universidad Católica de Chile

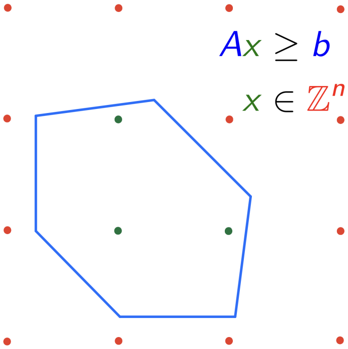
Motivation

- Symmetry is common: **almost 40% of problems** in the standard benchmark library for integer programming (MIPLIB) exhibit nontrivial symmetries.



Motivation

- It is folklore knowledge that **symmetry harms** state-of-the-art techniques for solving integer programs (IPs).



$Ax \geq b$

$x \in \mathbb{Z}^n$

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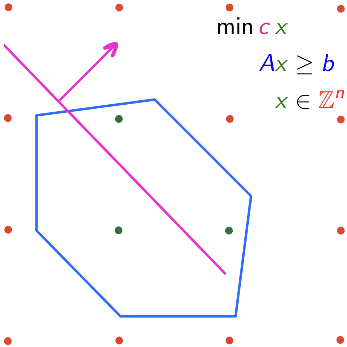

$$\begin{aligned} \min\ & c\,x \\ & Ax \geq b \\ & x \in \mathbb{Z}^n \end{aligned}$$

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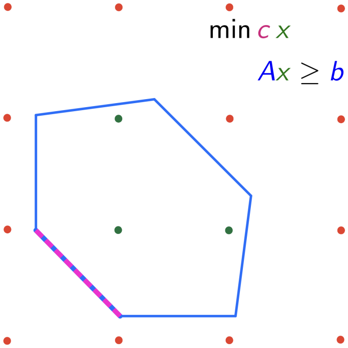

$$\min c\,x$$

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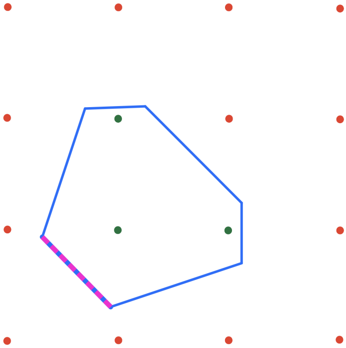

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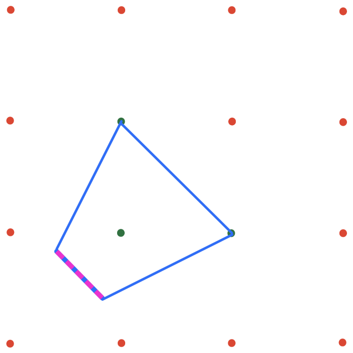
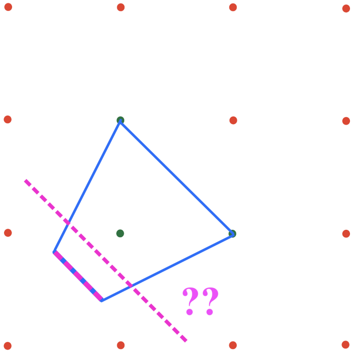


Motivation

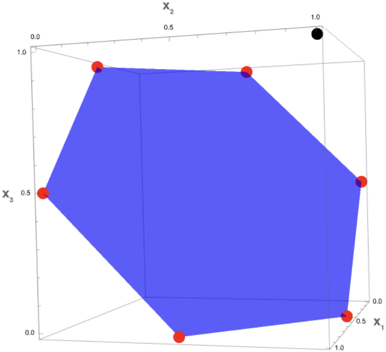
Goals

- Understand how **symmetries** affect the improvement of LP bounds of IPs under **Lift-and-Project** operators.
- Identify **key symmetry features** that characterize LP bound improvements, **beyond specific polytope classes.**



Symmetric IPs

$$\begin{aligned} \min \quad & x_1 + x_2 + x_3 \\ & x_1 - 2x_2 - 2x_3 \geq -3 \\ & -2x_1 + x_2 - 2x_3 \geq -3 \\ & -2x_1 - 2x_2 + x_3 \geq -3 \\ & 2x_1 + 2x_2 + 2x_3 \geq 3 \\ & x \in \{0,1\}^3 \end{aligned}$$



- Suppose $(\overline{x}_1, \overline{x}_2, \overline{x}_3)$ is feasible for the LP relaxation.
- Is $(\overline{x}_2, \overline{x}_1, \overline{x}_3)$ feasible?

Symmetric IPs

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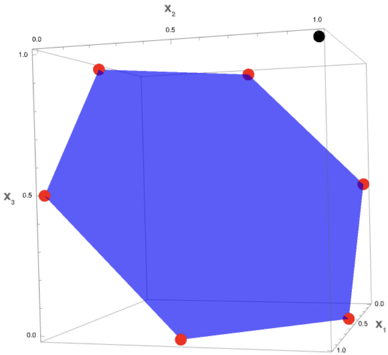


- Suppose $(\overline{x}_1, \overline{x}_2, \overline{x}_3)$ is feasible for the LP relaxation.
- Is $(\overline{x}_2, \overline{x}_1, \overline{x}_3)$ feasible? Yes!
- Any **permutation** of $(\overline{x}_1, \overline{x}_2, \overline{x}_3)$ is feasible.

Symmetries

Feasible permutations of $x = (x_1, \ldots, x_n)$ for every feasible x.

k-transitivity

Let $x \in P \subseteq [0,1]^5$.

1-transitive: $(i_1) \mapsto (j_1)$

$$(1) \mapsto (3)$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \rightsquigarrow \begin{pmatrix} x_5 \\ x_2 \\ x_1 \\ x_4 \\ x_3 \end{pmatrix} \in P$$

2-transitive: $(i_1, i_2) \mapsto (j_1, j_2)$

$$(2,5) \mapsto (3,1)$$

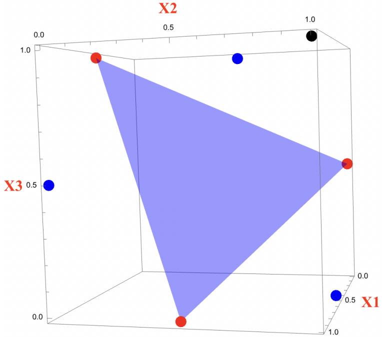
$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \rightsquigarrow \begin{pmatrix} x_5 \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in P$$

k-transitive polytope

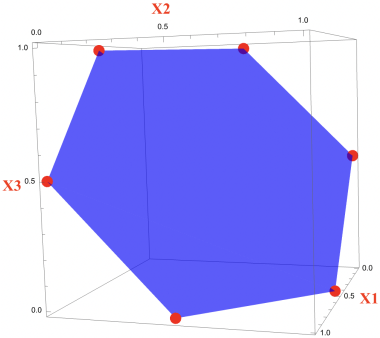
$P \subseteq [0,1]^n$ is k-**transitive** if $\forall x \in P$, and $\forall$ pair $(i_1, \ldots, i_k)$ and $(j_1, \ldots, j_k)$ of $[n]$, $\exists x' \in P$ permutation of x such that $\forall \ell \in [k]$,

$$x'_{j_\ell} = x_{i_\ell}.$$

k-transitivity


X2
0.0
0.5
1.0
1.0
X3 0.5
0.0
0.0
0.5 X1
1.0

conv$((1,1/2,0),(0,1,1/2),(1/2,0,1))$


X2
0.0
0.5
1.0
1.0
X3 0.5
0.0
0.0
0.5 X1
1.0

conv(all the permutations of $(1,1/2,0)$)

Lift-and-Project Operator

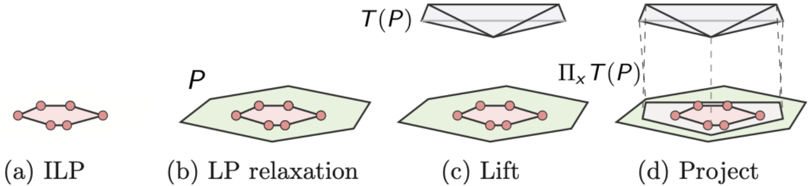

$T(P)$
P
$\Pi_x T(P)$
(a) ILP
(b) LP relaxation
(c) Lift
(d) Project

Figure 1: Fleming, Kothari, and Pitassi (2019)

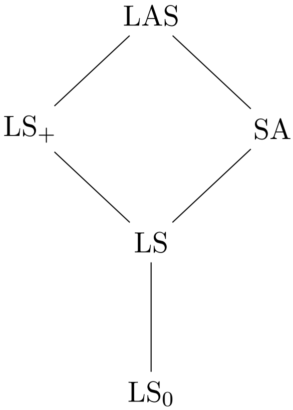
Definition

$T : \mathbb{R}^n \to \mathbb{R}^{n+r}$

- $T(P)$ is convex, $\Pi_x T(P) \subseteq P$;
- $\Pi_x T(P) \cap \{0,1\}^n = P \cap \{0,1\}^n$;
- $\forall S, R \subseteq [0,1]^n$ convex, $S \subseteq R \implies T(S) \subseteq T(R)$.

The Lift-and-Project Hierarchy

Seminal papers: Balas ('74), Sherali & Adams ('90), Lovász & Schrijver ('91), Balas, Ceria & Cornuéjols ('93), Lasserre ('01), Laurent ('03).



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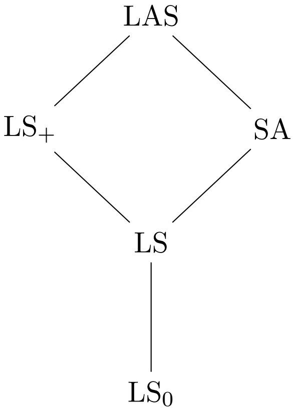

LAS

LS_+ SA

LS

LS_0

- The relaxations they produce can be vastly different.
- $T^k(\cdot)$ denotes the operator T "applied k times"
- In theory, $T^k(\cdot)$ can be computed in polynomial time for all these operators T and every constant k.
- In practice, e.g., computing LAS can be challenging already for $k = 2$.

First Theorem

Let $P \subseteq [0,1]^n$ be a $(k+1)$-transitive integer-empty polytope with $\dim P \leq n-1$. The following statements are equivalent:

1) P intersects all of the $(n-k)$-dimensional faces of $[0,1]^n$;
2) $\mathrm{SA}^k(P) \neq \emptyset$;
3) $\mathrm{LS}_+^k(P) \neq \emptyset$;
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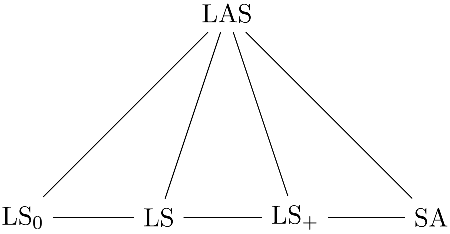

LAS

LS_0 — LS — LS_+ — SA

First Theorem: Example

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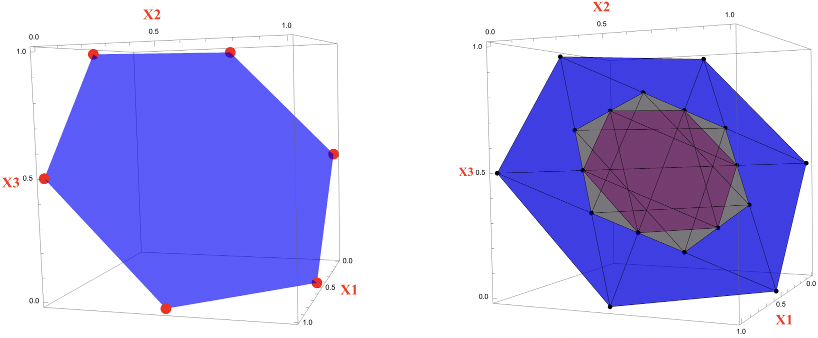
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0.0
0.0
0.5
X1
1.0

$P = \text{conv}(\text{all the permutations of } (1, 1/2, 0))$

Second Theorem ~~$\dim P \leq n - 1$~~

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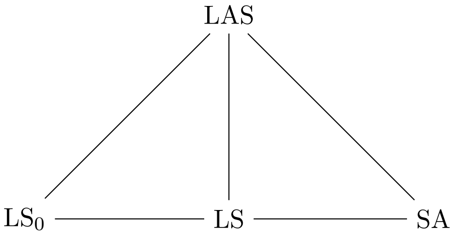


LAS

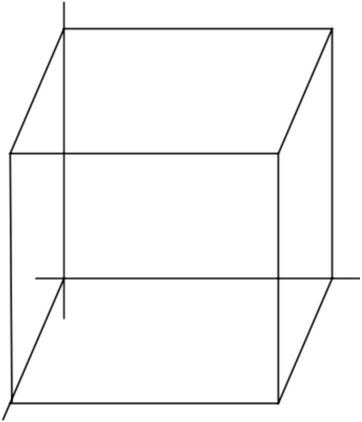
LS_0

LS

SA

The Cropped Cube

$$P^{cc} := \left\{ x \in [0,1]^n : \sum_{i \in I} x_i + \sum_{i \notin I} (1 - x_i) \geq 1/2, \ \forall I \subseteq [n] \right\}$$



The polytope P^{cc} is:

- integer-empty;
- *n*-transitive.

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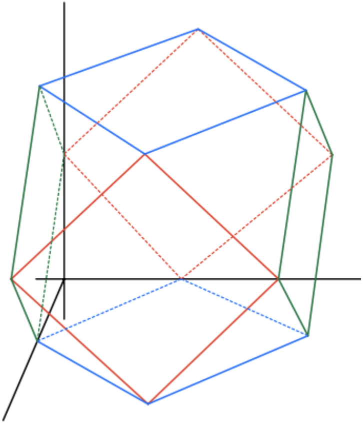

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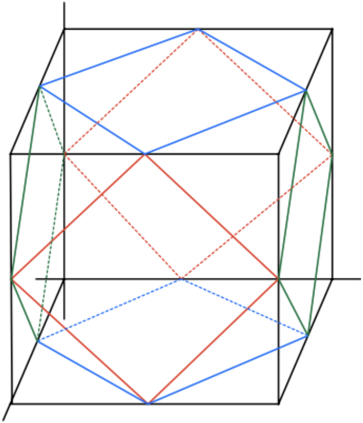

The polytope P^{cc} is:

- integer-empty;
- n-transitive.

- $\mathrm{LS}^{n-1}(P^{cc}) \neq \emptyset$ (Goemans & Tunçel, '01)
- $\mathrm{SA}^{n-1}(P^{cc}) \neq \emptyset$ (Laurent, '03)

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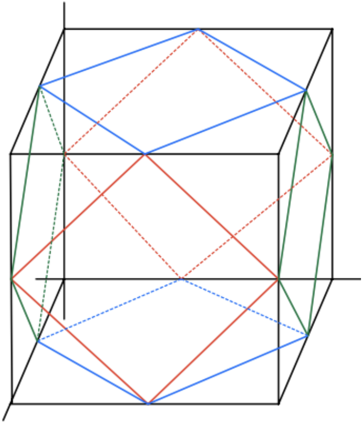

1) $P \cap F \neq \emptyset \ \forall F$ face of $[0,1]^n$ $\dim F = n - k$;
2) $\mathrm{SA}^k(P) \neq \emptyset$;
3) $\mathrm{LS}^k(P) \neq \emptyset$;
4) $\mathrm{LS}_0^k(P) \neq \emptyset$.

$\Rightarrow \mathrm{SA}^{n-1}(P^{cc}) \neq \emptyset$

Conclusions and Future Work

Key Takeaways

- Unified view of the strength of Lift-and-Project operators under symmetry.
- The more symmetric the optimal LP face, the harder to cut!
- “Hierarchy collapse” of Lift-and-Project operators under symmetry.

- Partial symmetries, i.e., group actions on a subset of variables.
- Other hierarchies, e.g., Lasserre.
- Other IP closures/cuts?

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